

PIR SENSOR GUIDE

Here’s a clear comparison of four PIR modules, focused on detection distance and the other practical details that matter.

	HC-SR501 Best overall when you need maximum range and full control (sensitivity + long adjustable delay). Biggest physical size.
	HC-SR505 Mini version of the classic. Fixed short delay (~8 s), shorter range, still needs 5 V-friendly supply.
	AM-312 Smallest and lowest power. Fixed ~2 s delay + 2 s lockout. Great for battery projects and tight spaces.
	HC-SR602 Very compact with the advantage of a widely adjustable delay (by changing a tiny resistor). Good middle ground between size and flexibility.

Detection Range (Distance) Overview

Model	Typical Range	Adjustable?	Notes
HC-SR501	3 – 7 m	Yes (pot)	Longest and most flexible. Turn sensitivity pot clockwise for more range.
HC-SR505	~3 m	No	Shorter, fixed range. Some listings claim up to 5–7 m, but real-world is usually around 3 m.
AM312	3 – 5 m	No	Compact version, typically solid to about 3–4 m in practice.
SR-602 (MH-SR602)	3 – 5 m (recommended 0–3.5 m)	Sensitivity via resistor change	Good mid-range for its tiny size.

Full Key Specs Comparison

Parameter	HC-SR501 (Standard)	HC-SR505	AM312	SR-602 (MH-SR602)
Detection Range	3–7 m (adj.)	~3 m	3–5 m	3–5 m (rec. ≤3.5 m)
Detection Angle	~110–120°	<100°	≤100°	~100–120°
Delay Time (HIGH)	5 s – 300 s (adj.)	~8 s ±30% (fixed)	~2 s (fixed)	2.5 s – 1 hour (adj. by resistor)
Blocking / Lockout	~2.5 s	~2–2.5 s	~2 s	~2 s (fixed)
Trigger Mode	H/L jumper (repeatable or single)	Repeatable (default)	Repeatable	Repeatable (fixed)
Operating Voltage	4.5–20 V	4.5–20 V	2.7–12 V	3.3–15 V
Standby Current	<50–65 µA	<60 µA	<20–100 µA (very low, often ~15–35 µA)	~20 µA
Output HIGH	3.3 V	3.3 V	~3.0–3.3 V	3.3 V
Size (approx.)	32 × 24 mm	10 × 23 mm	10 × 8 mm (smallest)	15–16 × 15–16 mm
Adjustable?	Yes (2 pots + jumper)	No	No	Delay yes (resistor), sensitivity limited

Detection Range Comparison (typical max)

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HC-SR501		7 m	(adjustable 3–7 m)
SR-602		5 m	(recommended ≤ 3.5 m)
AM312		5 m	(usually solid to ~3–4 m)
HC-SR505		3 m	(fixed)

Size Comparison (approximate PCB footprint)

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HC-SR501		32 × 24 mm	(largest)
HC-SR505		10 × 23 mm	
SR-602		~16 × 16 mm	
AM312		10 × 8 mm	(smallest)

Standby Power (lower is better for batteries)

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AM312		~15–35 µA	(best)
SR-602		~20 µA	
HC-SR501		~50–65 µA	
HC-SR505		~60 µA	

Additional Notes

